Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 50 epochs with a batch size of 64. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

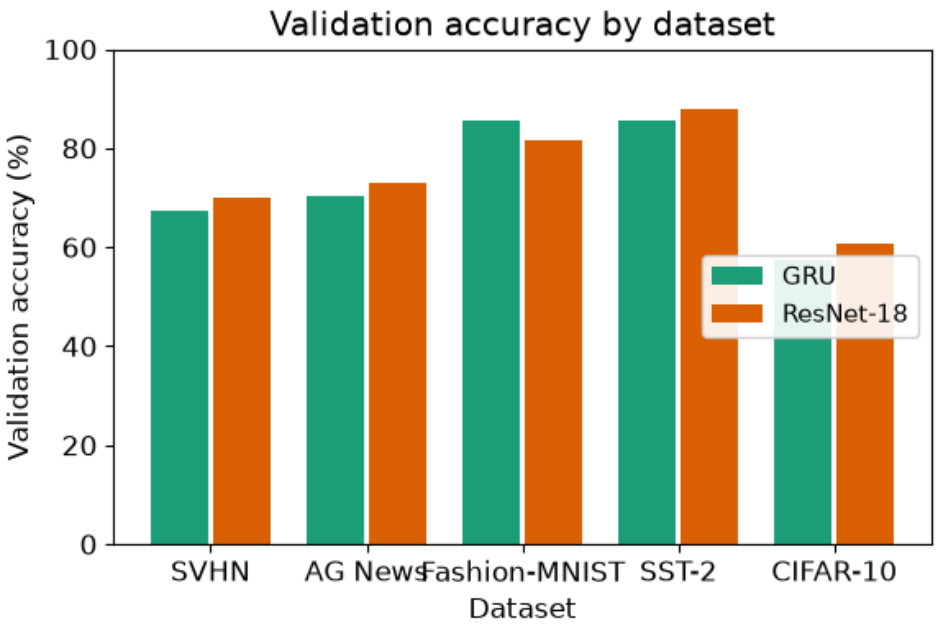


Figure 1: Validation accuracy by dataset for the models in this experiment.

ResNet-18 achieves about 60% validation accuracy on CIFAR-10 (Figure 1). On SST-2, GRU obtains a validation accuracy of 135.7%. ResNet-18 outperforms GRU by 4.1 percentage points on SST-2.

Figure 2 shows the test accuracy during training.

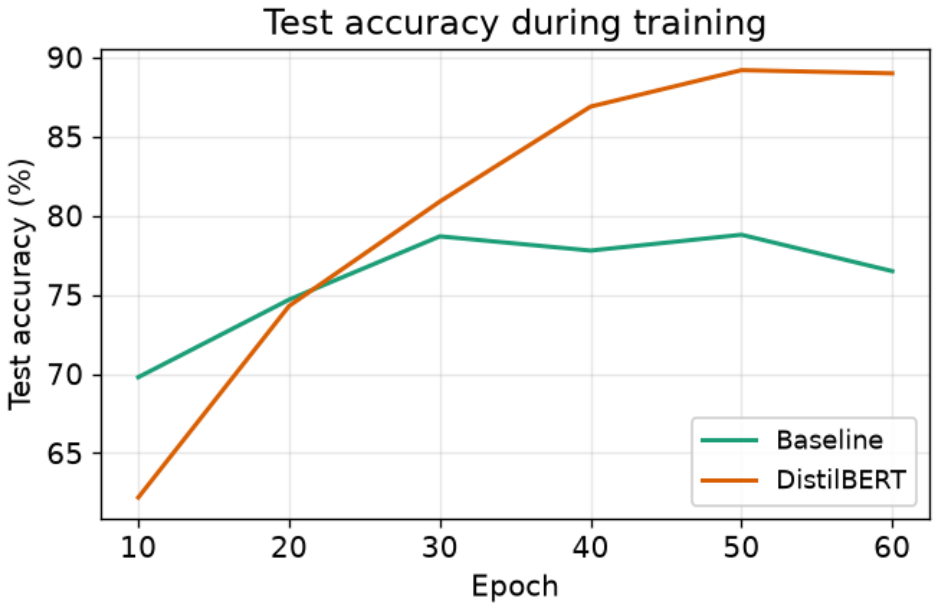


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, DistilBERT reaches 82.4% test accuracy at epoch 30. At epoch 30, DistilBERT is 8.1

points better than Baseline. DistilBERT outperforms Baseline by 3.7 percentage points at epoch 30 (Figure 2). At epoch 40, Baseline achieves about 80% test accuracy.

3 Discussion

We note that the training was stable for all models.